

API Reference

KMS to SDN Agent API - SDN server

API Version: 2.2.1

Messages the KMS sends to the SDN

CONTACT

URL: https://github.com/ait-crypto/kms-sdn-agent-interface-specification/discussions

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Security and Authentication

SECURITY SCHEMES

KEY	TYPE	DESCRIPTION
https	mutualTLS	Connections must use TLS 1.3. The server enforces this at the transport layer.

API

1. API

1.1 GET /version

Get this API version

State this SDN API version number which is implemented and supported by the KMS

REQUEST

No request parameters

RESPONSE

STATUS CODE - 200: Provided API version

RESPONSE MODEL - application/json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
version*	`string`	**PATTERN:**`^(0|[1-9]\d*)\.(0|[1-9]\d*)\.(0|[1-9]\d*)(?:-((?:0|[1-9]\d*|\d*[a-zA-Z-][0-9a-zA-Z-]*)(?:\.(?:0|[1-9]\d*|\d*[a-zA-Z-][0-9a-zA-Z-]*))*))?(?:\+([0-9a-zA-Z-]+(?:\.[0-9a-zA-Z-]+)*))?$` Version number in semantic versioning

STATUS CODE - 400: Bad request

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	`string`	Webpage with information on this problem
title*	`string`	Brief description of the problem
status*	`integer`	Response code or status code
detail*	`string`	Information on the problem and hints on how to solve it

STATUS CODE - 500: Internal error

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	`string`	Webpage with information on this problem
title*	`string`	Brief description of the problem
status*	`integer`	Response code or status code
detail*	`string`	Information on the problem and hints on how to solve it

2. LINK

2.1 GET /link/status

Get status of link

When an application queries the KMS for the status via ETSI GS QKD 014 /status message, the KMS has to provide the information how many keys are available on the link, but this involves remote peers, therefore the message is relied to the SDN to solve.

REQUEST

QUERY PARAMETERS

NAME	TYPE	EXAMPLE	DESCRIPTION
*master_sae_id	string	7184f8e9-4503-45d6-a399-afb9c0e61d06	Master SAE ID as per ETSI GS QKD 014
*slave_sae_id	string	9dd59892-db33-4db5-8e39-76b4a44e4aea	Slave SAE ID as per ETSI GS QKD 014

RESPONSE

STATUS CODE - 200: Successful path found and metrics available

RESPONSE MODEL - application/json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
source_kms_id*	string	Source KMS ID, most likely the one which received the query.
target_kms_id*	string	Remote peer KMS id.
key_size*	integer	>=0 Smallest number in bit given by KMS instances in the path stating their default key size, all other parameters use this for calculations.
stored_key_count*	integer	>=0 Currently available number of key entries which could be delivered to the application. Determined by the smallest number given by KMS instances in the path. Number of keys is based on the number of entries of the default key size.
max_key_count*	integer	>=0 Maximum possible key store in entries of default key sizes. Determined by the smallest number given by KMS instances in the path.
max_key_per_request*	integer	>=0 Maximum key batch allowed in one request. This is determined by the smallest number given by KMS instances in the path.
max_key_size*	integer	>=0 Maximum key size in bit, which KMS instances can handle. This is determined by the smallest number given by KMS instances in the path.
min_key_size*	integer	>=0 Minimum key size in bit, which can be handled. This is determined by the biggest number given by KMS instances in the path.

NAME	TYPE	DESCRIPTION
max_sae_id_count*	integer	>=0 Stating if group keys are supported and if so with how many additional participants. 0 means no additional SAE IDs. This is determined by the smallest number given by the specified source and destination KMS instances.
status_extension	object	

STATUS CODE - 400: Bad request

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

STATUS CODE - 500: Internal error

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

2.2 POST /link/registration

Post new link registration

When a source APP registers with a KMS, the KMS sends the parameters supplied by the APP to the SDN. The SDN configures a path between the source and destination, verifies the QoS and returns the key stream id and adjacent node information

REQUEST

REQUEST BODY - application/json

NAME	TYPE	DESCRIPTION
source*	string	URI or ID of the source APP
destination*	array	
qos_request*	object	
type*	enum	**ALLOWED:**etsi_004, etsi_014
etsi_014	object	
number*	integer	Number of keys
size*	integer	Size of keys in bit
extensions	array	
etsi_004	object	
key_chunk_size*	integer	length of key chunks in Byte.

NAME	TYPE	DESCRIPTION
max_bps*	integer	
min_bps*	integer	
jitter	integer	
priority	integer	
timeout	integer	
ttl*	integer	
metadata_mimetype	string	
performance*	array	
link_id	string	
current_kpi	{recursive}	kms-performance-entry
options	object	
multipath	boolean	In case of a multi-path establishment, specify 'true'.

RESPONSE

STATUS CODE - 201: Successful registration

RESPONSE MODEL - application/json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
key_stream_id*	string	Unique ID, corresponds to the ETSI 004 Key Stream ID parameter or identifies an ETSI 014 forwarding path
next_node_info*	array	
qos	object	

STATUS CODE - 400: Bad request

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

STATUS CODE - 500: Internal error

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

2.3 PATCH /link/registration

Update link status by destination APP

When the registered application at POST had keys successfully established, the PATCH is sent by the destination KMS. This indicates, that the path was closed.

REQUEST

REQUEST BODY - application/json

NAME	TYPE	DESCRIPTION
key_stream_id*	string	Unique ID, corresponds to the ETSI 004 Key Stream ID parameter or identifies an ETSI 014 forwarding path
destination*	array	

RESPONSE

STATUS CODE - 200: Successful registration

STATUS CODE - 400: Bad request

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

STATUS CODE - 404: Not found, API endpoint not found.

STATUS CODE - 500: Internal error

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

2.4 DELETE /link/{key_stream_id}

Delete key stream

If the ETSI 004 application closes a key stream, this is reported to the SDN

REQUEST

PATH PARAMETERS

NAME	TYPE	EXAMPLE	DESCRIPTION
*key_stream_id	uuid		Unique ID, corresponds to the ETSI 004 Key Stream ID parameter or identifies an ETSI 014 forwarding path

RESPONSE

STATUS CODE - 200: Link closed

STATUS CODE - 400: Bad request

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	`string`	Webpage with information on this problem
title*	`string`	Brief description of the problem
status*	`integer`	Response code or status code
detail*	`string`	Information on the problem and hints on how to solve it

STATUS CODE - 404: Not found, API endpoint not found.

STATUS CODE - 500: Internal error

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	`string`	Webpage with information on this problem
title*	`string`	Brief description of the problem
status*	`integer`	Response code or status code
detail*	`string`	Information on the problem and hints on how to solve it

3. NOTIFICATION

3.1 POST /notification/registration

Post KMS added to network

Inform the SDN of a new KMS that has been added to the network. If local instance has information about connected QKD devices or links, specify in request body.

REQUEST

REQUEST BODY - application/json

NAME	TYPE	DESCRIPTION
kms_id*	string	
model*	object	
device_vendor*	string	Vendor of the KMS
device_model*	string	Device brand name
device_version*	string	version number in semantic versioning
device_ids	array	
link_ids	array	

RESPONSE

STATUS CODE - 201: Registration received. Optionally additional config data provided in response. Config file data format or content not defined and expected to be vendor specific.

RESPONSE MODEL - application/octet-stream

STATUS CODE - 400: Bad request

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

STATUS CODE - 404: Not found, API endpoint not found

STATUS CODE - 500: Internal error

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

3.2 PATCH /notification/status

Update KMS status

Inform the SDN of when the KMS's status changes, e.g. the KMS goes into maintenance mode

REQUEST

REQUEST BODY - application/json

NAME	TYPE	DESCRIPTION
state*	`string`	
reason*	`string`	

RESPONSE

STATUS CODE - 200: Status received

STATUS CODE - 400: Bad request

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	`string`	Webpage with information on this problem
title*	`string`	Brief description of the problem
status*	`integer`	Response code or status code
detail*	`string`	Information on the problem and hints on how to solve it

STATUS CODE - 404: Not found, API endpoint not found

STATUS CODE - 500: Internal error

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	`string`	Webpage with information on this problem
title*	`string`	Brief description of the problem
status*	`integer`	Response code or status code
detail*	`string`	Information on the problem and hints on how to solve it

3.3 PUT /notification/event

Update event at KMS

Inform the SDN of an event

REQUEST

REQUEST BODY - application/json

NAME	TYPE	DESCRIPTION
level*	enum	ALLOWED:debug, info, warning, error, critical
code	integer	
reason*	string	

RESPONSE

STATUS CODE - 200: Event information received

STATUS CODE - 400: Bad request

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it

STATUS CODE - 404: Not found, API endpoint not found.

STATUS CODE - 500: Internal error

RESPONSE MODEL - application/problem+json

NAME	TYPE	DESCRIPTION
OBJECT WITH BELOW STRUCTURE		
type	string	Webpage with information on this problem
title*	string	Brief description of the problem
status*	integer	Response code or status code
detail*	string	Information on the problem and hints on how to solve it